

CAF — Operations & Deploy

Aggregated export — 2026-07-10. Source markdown in repo is unchanged.

Bundle id: `09-caf-operations-and-deploy`

Included: docs/FLY_PRODUCTION_CHECKLIST.md

Goal

Reduce time-to-complete and eliminate common Fly.io failure modes (timeouts, cold starts, wrong base URLs, renderer/video-assembly hangs) for **CAF Core** + **CAF Renderer/Media Gateway** deployments.

Operator Review / Admin UI: <https://caf-core.fly.dev/admin/workbench> (embedded `apps/review` in the Core image). Deploy with `fly deploy -a caf-core` from repo root — not Vercel.

Biggest latency contributors (where to look)

- **LLM generation (OpenAI chat):** `src/services/openai-chat.ts` , `src/services/llm-generator.ts`
- **Carousel rendering** (per-slide Puppeteer): `services/renderer/server.js` + Core caller in `src/services/job-pipeline.ts` (`POST /render-binary` per slide)
- **Scene video assembly:**
 - **Sora polling:** `src/services/sora-scene-clips.ts` (OpenAI Videos API polling + download/upload)
 - **HeyGen polling:** `src/services/heygen-renderer.ts`
 - **Concat/mux:** `services/video-assembly/server.js` (ffmpeg)

Fly deployment shapes (pick one)

Option A (recommended): combined gateway on one Fly app

Deploy `services/media-gateway` image (spawns renderer + video-assembly on localhost).

- **Pros:** one stable public base URL for both render + video endpoints; avoids “wrong base URL” errors; easiest for `RENDERER_BASE_URL` / `VIDEO_ASSEMBLY_BASE_URL` .

- **Deploy:** `fly deploy -c fly.caf-renderer.toml`
- **Public endpoints:**
 - Renderer: `POST /render-binary` , `GET /renderer/ready` , `GET /renderer/health` , `GET /templates/*`
 - Video assembly: `POST /concat-videos` , `POST /mux` , `GET /status/:id` , `POST /full-pipeline`
 - Gateway readiness: `GET /ready`

Option B: standalone services (only if you must)

- **services/renderer** Fly app: Puppeteer only.
- **services/video-assembly** Fly app: ffmpeg only.
- **Downside:** you must set two different base URLs and ensure the "status polling" endpoint is exposed on the same host you use for async jobs.

Required env vars (exact)

On `caf-core` (Fly app `caf-core`)

- **RENDERER_BASE_URL :**
 - If using combined gateway: `https://caf-renderer.fly.dev`
 - If using standalone renderer: `https://<your-renderer-app>.fly.dev`
- **VIDEO_ASSEMBLY_BASE_URL :**
 - If using combined gateway: `https://caf-renderer.fly.dev`
 - If using standalone video-assembly: `https://<your-video-assembly-app>.fly.dev`

Constraints / gotchas:

- **No trailing slash** is safest (Core trims it, but keep it consistent).
- **VIDEO_ASSEMBLY_BASE_URL** must serve **both** `POST /concat-videos?async=1` and `GET /status/:requestId` . If you point it at a host that only proxies the POST routes, Core will fail during polling.

On `caf-renderer` (combined gateway app)

- **Recommended:**
 - `CAF_TEMPLATE_API_URL=https://caf-core.fly.dev` (so renderer can fetch missing templates from Core)
 - `SUPABASE_URL` , `SUPABASE_SERVICE_ROLE_KEY` , `SUPABASE_ASSETS_BUCKET` (so video-assembly can upload outputs and

```
return public_url )
```

- **Timeout hardening** (safe production defaults):
 - VIDEO_ASSEMBLY_FETCH_TIMEOUT_MS=180000
 - VIDEO_ASSEMBLY_FFMPEG_TIMEOUT_MS=1200000
 - VIDEO_ASSEMBLY_JOB_TIMEOUT_MS=1800000

OpenAI timeout hardening (CAF Core)

- OPENAI_CHAT_TIMEOUT_MS=180000
- OPENAI_CHAT_MAX_RETRIES=2

Notes:

- These are used by `src/services/openai-chat.ts` and `src/services/openai-chat-multimodal.ts` to prevent “hang forever” calls.

Fly health checks (readiness vs liveness)

- **caf-core**: Fly should check `GET /readyz` (DB connectivity)
- **caf-renderer (combined gateway)**: Fly should check `GET /ready` (Chromium launch + ffmpeg present)

Observability (minimal, production-safe)

- **Video-assembly** logs durations for `stitch`, `concat`, `mux`, and full-pipeline runs.
- **OpenAI chat** emits a console log when a call takes ($\geq$ 20s), including attempt number and tokens.
- **Core video render** emits a console log when the overall render step takes ($\geq$ 20s).

Fly smoke test plan

1) Readiness

- `curl https://caf-core.fly.dev/readyz`
- `curl https://caf-renderer.fly.dev/ready`

2) Dependency wiring sanity

- `curl https://caf-core.fly.dev/health/rendering`
 - Confirm `rendering.renderer.ok=true`
 - Confirm `rendering.video_assembly.ok=true`

3) Minimal renderer test

- ```
curl -X POST https://caf-renderer.fly.dev/render-binary -H
'content-type: application/json' -d
'{"template":"carousel_sns_chat_story","data":
{"render":{"slides":
[{"headline":"Hello","body":"World"}]}}' --output
out.png
```

### 4) Minimal video-assembly async test (concat)

- Start:
  - ```
curl -X POST "https://caf-renderer.fly.dev/concat-
videos?async=1" -H "content-type: application/json"
-d '{"video_urls":["<public-mp4-1>","<public-mp4-
2>"]}'
```
- Poll:
 - ```
curl https://caf-
renderer.fly.dev/status/<request_id>
```

### 5) End-to-end Core pipeline (small)

- Create a run with 1 candidate and process it (or use existing admin helpers).
- Watch logs:
  - Core logs for OpenAI timeouts/retries and overall video render durations.
  - Renderer/gateway logs for /ready readiness and any puppeteer errors.
  - Video-assembly logs for ffmpeg duration and any watchdog timeouts.

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## Included: docs/USER\_INPUT\_AND\_SECRETS.md

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## Everything you need to provide (save this checklist)

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Use this as your **single vault checklist**. Copy values only into `.env`, password managers, or deployment secret stores (Vercel / Fly) — never into git.

For variable names and comments, see also [.env.example](#) and [ENV\\_AND\\_SECRETS\\_INVENTORY.md](#).

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### 1. CAF Core API (this service)

| Item | Required? | Where it | Notes |
|------|-----------|----------|-------|
|------|-----------|----------|-------|

|                                                                         |                       | goes                    |                                                                                                                                           |
|-------------------------------------------------------------------------|-----------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Postgres</b> DATABASE_URL                                            | <b>Yes</b>            | .env /<br>Fly<br>secret | Supabase pooler URI, Neon, RDS, or local Docker (see docker-compose.yml).                                                                 |
| CAF_CORE_API_TOKEN                                                      | Recommended prod      | .env /<br>Fly<br>secret | Long random string; pair with <b>CAF_CORE_REQUIRE_AUTH=1</b> .                                                                            |
| PORT / HOST                                                             | Optional              | .env                    | Defaults 3847 / 0.0.0.0.                                                                                                                  |
| OPENAI_API_KEY / OPENAI_MODEL                                           | Yes for generation    | .env /<br>Fly<br>secret | Required for LLM generation, scene clips (Sora), TTS.                                                                                     |
| HEYGEN_API_KEY / HEYGEN_API_BASE                                        | Yes for HeyGen video  | .env /<br>Fly<br>secret | Only if using HeyGen avatar / video-agent flows.                                                                                          |
| SUPABASE_URL /<br>SUPABASE_SERVICE_ROLE_KEY /<br>SUPABASE_ASSETS_BUCKET | Yes for asset uploads | .env /<br>Fly<br>secret | Where render outputs (carousel PNGs, video MP4s, scene clips) are stored.                                                                 |
| RENDERER_BASE_URL /<br>VIDEO_ASSEMBLY_BASE_URL                          | Yes for rendering     | .env /<br>Fly<br>secret | Point at services/renderer, services/video-assembly, or the combined media-gateway.                                                       |
| CAROUSEL_TEMPLATES_DIR                                                  | Optional              | .env                    | Directory served via GET /api/templates/* for the renderer.                                                                               |
| CAF_PUBLISH_EXECUTOR                                                    | Optional              | .env                    | none (default) / dry_run / meta. For meta, add page tokens + <b>META_GRAPH_API_VERSION</b> (or per-project rows in project_integrations). |
| CAF_OUTPUT_SCHEMA_VALIDATION_MODE                                       | Optional              | .env                    | skip / warn / enforce; legacy <b>CAF_SKIP_OUTPUT_SCHEMA_VALIDATION</b> is still honored as a fallback.                                    |
| Scoring & caps (SCORE_WEIGHT_*,<br>DEFAULT_MIN_SCORE_TO_GENERATE, ...)  | Optional              | .env                    | Tuning only. See src/config.ts.                                                                                                           |

## Migrations:

- Applied via **npm run migrate**, or automatically on API startup when **CAF\_RUN\_MIGRATIONS\_ON\_START** is true (default).
- Tracked in **caf\_core.schema\_migrations**.
- If you already applied migrations before the ledger existed, baseline it once:

```
INSERT INTO caf_core.schema_migrations (filename) VALUES
('001_caf_core_schema.sql')
ON CONFLICT DO NOTHING;
```

## 2. Review app ( apps/review — Next.js 14)

The Review workbench is a **pure client of CAF Core**. It does not own its own database, queue, or third-party credentials — everything it shows comes from Core `/v1/*` routes.

| Item                         | Required?                                              | Notes                                                                                                                               |
|------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| CAF_CORE_URL                 | <b>Yes</b>                                             | Public URL of your Core deployment (e.g. <code>https://caf-core.fly.dev</code> ); <code>http://localhost:3847</code> for local dev. |
| CAF_CORE_TOKEN               | Yes when Core has <code>CAF_CORE_REQUIRE_AUTH=1</code> | Must match <code>CAF_CORE_API_TOKEN</code> on Core. Sent as <code>x-caf-core-token</code> / <code>Authorization: Bearer</code> .    |
| REVIEW_WRITE_TOKEN           | Recommended prod                                       | Gate decision writes from the UI; leave empty for local dev.                                                                        |
| NEXT_PUBLIC_APP_URL          | Recommended                                            | Used for preview/link building.                                                                                                     |
| PROJECT_SLUG                 | Optional                                               | Empty = load all active projects' queues. Set (e.g. <code>SNS</code> , <code>connectr</code> ) to lock the workbench to one tenant. |
| REVIEW_ALL_PROJECTS          | Optional                                               | <code>1</code> / <code>true</code> forces cross-project queue even when <code>PROJECT_SLUG</code> is set.                           |
| REVIEW_FALLBACK_PROJECT_SLUG | Optional                                               | Used only when Core lacks <code>/v1/review-queue-all/*</code> (older deploy). Default fallback is <code>SNS</code> .                |

All Core-dependent configuration (HeyGen, Supabase, Meta publishing, Google/OAuth credentials) stays on **Core** and `project_integrations` — not on the Review app.

## 3. Media services on Fly.io

| Item          | Notes                                                                                |
|---------------|--------------------------------------------------------------------------------------|
| Fly app names | e.g. <code>caf-core</code> , <code>media-gateway</code> (renderer + video-assembly). |

|                                                                              |                                                                                                 |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>FLY_API_TOKEN</code>                                                   | GitHub Actions secret for the deploy workflow.                                                  |
| <code>SUPABASE_URL</code> +<br><code>SUPABASE_SERVICE_ROLE_KEY</code> on Fly | Same Supabase project as Core uploads.                                                          |
| <code>OPENAI_API_KEY</code> on Fly (if needed)                               | Only when the video-assembly uses Whisper / mux alignment.                                      |
| <code>CAF_TEMPLATE_API_URL</code>                                            | Core base URL, so the renderer can fetch .hbs templates via <code>GET /api/templates/*</code> . |
| Renderer / mux tuning                                                        | Timeouts, burn style, retries — listed in <code>.env.example</code> .                           |

See also [FLY\\_PRODUCTION\\_CHECKLIST.md](#) .

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## 4. Third-party provider accounts (vault)

| Provider          | Typical secret                                       | Used by                                                       |
|-------------------|------------------------------------------------------|---------------------------------------------------------------|
| <b>OpenAI</b>     | API key (+ org / project id if scoped)               | LLM generation, Sora scene clips, TTS, approval-review vision |
| <b>HeyGen</b>     | API key / account                                    | Avatar and video-agent renders                                |
| <b>Supabase</b>   | Service-role key                                     | Asset storage bucket                                          |
| <b>Meta Graph</b> | Page tokens (per <code>project_integrations</code> ) | Publishing when <code>CAF_PUBLISH_EXECUTOR=meta</code>        |

Per-project overrides (e.g. separate Meta tokens per brand) live in `caf_core.project_integrations` , not in env.

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## 5. External publish workers (optional)

If you run a downstream worker to actually post content, it consumes Core's stable publish payload:

- **Endpoint:** `GET /v1/publications/:project_slug/:id/n8n-payload` (route name is historical — the shape is the Core-owned contract for any external publisher).
- **Auth:** `x-caf-core-token` / `Authorization: Bearer` when `CAF_CORE_REQUIRE_AUTH=1` .
- **Outcomes:** workers call back into `POST /v1/publications/.../complete` (or patch the placement) to record `published` / `failed` state.

Nothing in Core requires a specific external orchestrator; the publish executor is selectable via `CAF_PUBLISH_EXECUTOR` ( `none` / `dry_run` / `meta` ).

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## 6. End-state "done when" checklist

- ☐ `DATABASE_URL` set; `npm run migrate` succeeds; `npm run dev` responds on `/health` .
- ☐ If auth on: `CAF_CORE_REQUIRE_AUTH=1` + `CAF_CORE_API_TOKEN` ; every client (Review app, scripts, workers) sends the matching token.
- ☐ Review app deployed with `CAF_CORE_URL` (public) + optional `CAF_CORE_TOKEN` ; review queue loads.
- ☐ Fly apps deployed; Core's `RENDERER_BASE_URL` / `VIDEO_ASSEMBLY_BASE_URL` point at the deployed media services.
- ☐ `OPENAI_API_KEY` , `HEYGEN_API_KEY` , and `SUPABASE_*` set wherever they are needed (Core, and on Fly for the media stack that uploads assets).
- ☐ `CAF_PUBLISH_EXECUTOR` set explicitly for production ( `meta` with project-integration tokens, or `none` / `dry_run` while testing).
- ☐ No secrets committed; `.env` is git-ignored.

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*This file is safe to commit. It contains no secret values.*

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## Included: ENV\_AND\_SECRETS\_INVENTORY.md

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### Environment variables and secrets inventory (CAF)

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**Purpose:** One checklist of **every** configuration name used by this repo and adjacent ops (n8n, Fly, Vercel), where to set it, and whether it is secret.

**External readers:** See also `docs/EXTERNAL_CONTEXT_PACK.md` (Tier 1 includes this file + `.env.example` ) and `docs/REBUILD_FROM_DOCS.md` (minimum env for bootstrap).

#### Security — non-negotiable

- **Never** commit real tokens, keys, JSON service accounts, or refresh tokens to git.
  - Copy `.env.example` → `.env` locally ( `.env` is gitignored).
  - Store production values in **Vercel / Fly secrets / n8n credentials / your vault** only.
  - The “**Your value (vault only)**” column below is for you to fill **outside** the repo.
- 

## Legend



| Column              | Meaning                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Secret?</b>      | yes = treat like a password; no = non-sensitive URL or flag; optional secret = only if you use that feature |
| <b>Where to set</b> | Typical location for production                                                                             |

## 0. CAF Core API (repo root) — `src/config.ts`

**Canonical list:** every variable is defined in `src/config.ts` (Zod) with defaults. Comments in `.env.example`.

| Variable                                                        | Secret?         | Required         | Purpose                                      |
|-----------------------------------------------------------------|-----------------|------------------|----------------------------------------------|
| DATABASE_URL                                                    | <b>yes</b>      | <b>yes</b>       | PostgreSQL connection (caf_core schema)      |
| OPENAI_API_KEY                                                  | <b>yes</b>      | for LLM          | Generation, QC helpers, mimic copy, insights |
| OPENAI_MODEL                                                    | no              | recommended      | Default chat model                           |
| PORT / HOST                                                     | no              | optional         | API listen (default <b>3847</b> )            |
| CAF_CORE_REQUIRE_AUTH                                           | no              | optional         | 1 to require token on protected routes       |
| CAF_CORE_API_TOKEN                                              | <b>yes</b>      | when auth on     | x-caf-core-token Or Authorization: Bearer    |
| CAF_RUN_MIGRATIONS_ON_START                                     | no              | optional         | Apply migrations on startup (default on)     |
| RENDERER_BASE_URL                                               | no              | for carousel     | Carousel worker URL                          |
| VIDEO_ASSEMBLY_BASE_URL                                         | no              | for video        | ffmpeg worker URL                            |
| SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, SUPABASE_ASSETS_BUCKET | <b>yes</b> / no | for assets       | Object storage uploads                       |
| HEYGEN_API_KEY                                                  | <b>yes</b>      | for HeyGen flows | Video agent / avatar renders                 |
| MIMIC_IMAGE_ENABLED                                             | no              | optional         | Master switch for mimic render               |
| MIMIC_IMAGE_PROVIDER                                            | no              | optional         | bfl   dashscope   nvidia   openai            |

|                                                    |     |              |                                    |
|----------------------------------------------------|-----|--------------|------------------------------------|
| BFL_API_KEY, DASHSCOPE_API_KEY, NVIDIA_NIM_API_KEY | yes | per provider | Mimic image edit providers         |
| CAF_PUBLISH_EXECUTOR                               | no  | optional     | none   dry_run   meta              |
| CAF_REQUIRE_HUMAN_REVIEW_AFTER_QC                  | no  | optional     | Human gate after QC (default true) |
| CAF_OUTPUT_SCHEMA_VALIDATION_MODE                  | no  | optional     | skip   warn   enforce              |

Mimic-specific vars: **\$10** below. Full bootstrap: [docs/REBUILD\\_FROM\\_DOCS.md](#) .

## 1. Review app ( apps/review ) — Vercel / local

Review is a **Next.js client of CAF Core** — job state lives in Postgres via Core APIs, not in the Review app database.

| Variable                     | Secret? | Required          | Where to set                       | Used for                                             |
|------------------------------|---------|-------------------|------------------------------------|------------------------------------------------------|
| CAF_CORE_URL                 | no      | yes               | Vercel env, apps/review/.env.local | Core API base<br>http://localhost                    |
| CAF_CORE_TOKEN               | yes     | when Core auth on | Vercel env                         | Sent to Core as core-token                           |
| CAF_CORE_API_TOKEN           | yes     | optional alias    | Some routes                        | Alternate name by a few API routes                   |
| REVIEW_WRITE_TOKEN           | yes     | optional          | Vercel env                         | Protects POST /api/task/.../ in Review               |
| RENDERER_BASE_URL            | no      | recommended       | Vercel env                         | Carousel preview proxies (default http://localhost)  |
| NEXT_PUBLIC_APP_URL          | no      | recommended       | Vercel env                         | Absolute links                                       |
| PROJECT_SLUG                 | no      | optional          | Vercel env                         | Lock workbench one tenant; error all active projects |
| REVIEW_ALL_PROJECTS          | no      | optional          | Vercel env                         | 1/true = cross-queue even if PROJECT_SLUG S          |
| REVIEW_FALLBACK_PROJECT_SLUG | no      | optional          | Vercel env                         | Fallback when Core lacks /v1, queue-all/*            |

**Legacy (not used by current Core-backed Review):** Google Sheets queue vars ( `GOOGLE_REVIEW_QUEUE_*` , `GOOGLE_SERVICE_ACCOUNT_JSON` , etc.) may appear in old deploy notes — the current app reads the review queue from **Core /v1/ APIs**.

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## 2. Renderer service ( `services/renderer` ) — Fly or local

| Variable                 | Secret?    | Required | Where to set | Used for                                                       | Your value (vault only) |
|--------------------------|------------|----------|--------------|----------------------------------------------------------------|-------------------------|
| PORT                     | no         | optional | Fly, local   | Listen (default <b>3333</b> )                                  |                         |
| CAF_TEMPLATE_API_URL     | no         | optional | Fly secret   | Base URL of <b>this</b> Next app for Supabase-backed templates |                         |
| RENDERERS_BEFORE_RESET   | no         | optional | Fly          | Puppeteer restarts after N renders (default 12)                |                         |
| RENDERER_SHUTDOWN_SECRET | <b>yes</b> | optional | Fly secret   | POST /shutdown protection                                      |                         |

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## 3. Video assembly ( `services/video-assembly` ) — Fly or local

| Variable                                 | Secret?    | Required | Where to set | Used for                        | Your value (vault only) |
|------------------------------------------|------------|----------|--------------|---------------------------------|-------------------------|
| PORT                                     | no         | optional | Fly, local   | Listen (default <b>3334</b> )   |                         |
| SUPABASE_URL or NEXT_PUBLIC_SUPABASE_URL | no         | yes      | Fly secret   | Upload target project           |                         |
| SUPABASE_SERVICE_ROLE_KEY                | <b>yes</b> | yes      | Fly secret   | Storage + optional tasks update |                         |
| SUPABASE_ASSETS_BUCKET                   | no         | optional | Fly secret   | Bucket (default assets)         |                         |

|                                                 |            |          |               |                                           |  |
|-------------------------------------------------|------------|----------|---------------|-------------------------------------------|--|
| OPENAI_API_KEY                                  | <b>yes</b> | optional | Fly<br>secret | Whisper<br>alignment<br>when<br>requested |  |
| OPENAI_WHISPER_MODEL                            | no         | optional | Fly           | Default<br>whisper-1                      |  |
| OPENAI_WHISPER_TIMEOUT_MS                       | no         | optional | Fly           | Whisper HTTP<br>timeout                   |  |
| VIDEO_ASSEMBLY_FETCH_TIMEOUT_MS                 | no         | optional | Fly           | Download<br>clips/video                   |  |
| VIDEO_ASSEMBLY_FFMPEG_TIMEOUT_MS                | no         | optional | Fly           | ffmpeg<br>processes                       |  |
| VIDEO_ASSEMBLY_JOB_TIMEOUT_MS                   | no         | optional | Fly           | Whole<br>stitch/mux<br>job                |  |
| FFMPEG_PATH                                     | no         | optional | Fly           | Override<br>binary                        |  |
| FFPROBE_PATH / FFPROBE_TIMEOUT_MS               | no         | optional | Fly           | Probe media                               |  |
| TMP_DIR                                         | no         | optional | Fly           | Temp working<br>dir                       |  |
| MUX_BURN_SUBTITLE_FORCE_STYLE                   | no         | optional | Fly           | ASS<br>force_style                        |  |
| MUX_BURN_ENCODE_PRESET /<br>MUX_BURN_ENCODE_CRF | no         | optional | Fly           | Burn-in<br>encode                         |  |
| MUX_CHUNK_SCRIPT_MAX_CHARS                      | no         | optional | Fly /<br>body | TTS chunking                              |  |
| MUX_MIN_CAPTION_SEC                             | no         | optional | Fly           | Caption<br>pacing                         |  |
| MUX_CAPTION_WEIGHT_EXPONENT                     | no         | optional | Fly           | Caption<br>pacing                         |  |
| MUX_CAPTION_TIMING_MODE                         | no         | optional | Fly           | equal /<br>weighted                       |  |
| MUX_USE_SCENE_DURATION_FOR_SRT                  | no         | optional | Fly           | Legacy timing                             |  |
| MUX_SUBTITLE_DURATION_SCALE                     | no         | optional | Fly           | Scale scene<br>SRT                        |  |
| MUX_SUBTITLE_ALIGN                              | no         | optional | Fly           | e.g. whisper                              |  |

|                                                            |    |          |     |                           |  |
|------------------------------------------------------------|----|----------|-----|---------------------------|--|
| MUX_WHISPER_CAPTION_SOURCE                                 | no | optional | Fly | payload / transcript      |  |
| MUX_SYNC_SUPABASE_TASK                                     | no | optional | Fly | 0 to skip task row update |  |
| SUPABASE_UPLOAD_MAX_ATTEMPTS /<br>SUPABASE_UPLOAD_RETRY_MS | no | optional | Fly | Upload retry              |  |

## 4. Media gateway ( `services/media-gateway` ) — Fly

| Variable            | Secret? | Required | Where to set | Used for                              | Your value (vault only) |
|---------------------|---------|----------|--------------|---------------------------------------|-------------------------|
| PORT                | no      | optional | Fly          | Gateway listen (default <b>8080</b> ) |                         |
| RENDERER_PORT       | no      | optional | Fly          | Child renderer port                   |                         |
| VIDEO_ASSEMBLY_PORT | no      | optional | Fly          | Child video-assembly port             |                         |

Child processes inherit **same env** as gateway on Fly — set Supabase and mux vars on the **gateway app** so stitch/mux can upload.

## 5. GitHub Actions (deploy Fly media-gateway)

| Name             | Type                       | Secret?    | Where to set                  | Used for                                                          | Your value (vault only) |
|------------------|----------------------------|------------|-------------------------------|-------------------------------------------------------------------|-------------------------|
| FLY_API_TOKEN    | repository <b>Secret</b>   | <b>yes</b> | GitHub → Settings → Secrets   | flyctl deploy ( <code>.github/workflows/fly-renderer.yml</code> ) |                         |
| FLY_RENDERER_APP | repository <b>Variable</b> | no         | GitHub → Settings → Variables | Override Fly app name (-a)                                        |                         |

## 6. n8n (instance credentials — not in `.env.example` )

Store these in **n8n** → **Credentials**, not in the CAF repo.

| Credential type           | Typical use in CAF flows                                   | Your value (vault only) |
|---------------------------|------------------------------------------------------------|-------------------------|
| Google Sheets OAuth2      | CREATION, VALIDATION, INPUT, PROCESSING, Signal Pack       |                         |
| OpenAI API                | LLM, TTS, Sora/video nodes                                 |                         |
| Supabase                  | Direct DB/REST if any node uses it                         |                         |
| HeyGen                    | Video render                                               |                         |
| HTTP Header Auth / Bearer | RENDERER_BASE_URL, VIDEO_ASSEMBLY_BASE_URL, mux async poll |                         |
| Custom                    | Apify, webhooks, etc.                                      |                         |

Also record:

| Item                                | Your value (vault only) |
|-------------------------------------|-------------------------|
| n8n base URL                        |                         |
| n8n admin auth method               |                         |
| Per-workflow webhook URLs (if used) |                         |

## 7. Google Cloud / Sheets (operational inventory)

Fill in **outside** the repo (which spreadsheet belongs to which env).

| Resource                                   | Example / note                     | Your value (vault only) |
|--------------------------------------------|------------------------------------|-------------------------|
| GCP project for Sheets API                 |                                    |                         |
| OAuth consent / Desktop or Web client      |                                    |                         |
| Service account email (if used)            | Share each spreadsheet with Editor |                         |
| <b>VALIDATION</b> spreadsheet ID           | Review Queue tab                   |                         |
| <b>CREATION - Runtime</b> spreadsheet ID   | Signal_Packs, Content_Jobs, ...    |                         |
| <b>PROCESSING</b> insights spreadsheet ID  | Write Signal Pack reads from here  |                         |
| <b>Flow Engine</b> spreadsheet ID          | Global                             |                         |
| <b>CREATION - Project Config</b> per brand |                                    |                         |

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## 8. Other platforms (from 09\_external\_integrations.md )

| Platform | What to store                                                                    | Your value (vault only) |
|----------|----------------------------------------------------------------------------------|-------------------------|
| Supabase | Project ref, dashboard URL, service role key (above), anon key if any client use |                         |
| Fly.io   | Org, app name(s) for gateway/renderer, region                                    |                         |
| Vercel   | Project name, team, env vars (mirror \$1)                                        |                         |
| OpenAI   | Org / project IDs if using scoped keys                                           |                         |
| HeyGen   | API key, account                                                                 |                         |
| Apify    | API token, actor IDs                                                             |                         |

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## 9. Quick “first deploy” checklist

- ☐ Supabase: project created; migrations or SQL applied; bucket `assets` public as needed
  - ☐ Vercel: \$1 vars; `NEXT_PUBLIC_APP_URL` = production URL
  - ☐ Google: Review Queue sheet shared with SA or OAuth account (Editor)
  - ☐ Fly (optional): deploy media-gateway; set Supabase + mux + optional `CAF_TEMPLATE_API_URL`
  - ☐ GitHub: `FLY_API_TOKEN` (+ optional `FLY_RENDERER_APP` variable)
  - ☐ n8n: credentials + `RENDERER_BASE_URL` / `VIDEO_ASSEMBLY_BASE_URL` pointing at stable URLs
  - ☐ Rotate any credential that ever touched a public repo
- 

## 10. CAF Core API — top-performer mimic ( `src/config.ts` )

Set on the **Core** host (repo root `.env` / Fly secrets). Copy generation requires `OPENAI_API_KEY` even when render uses BFL/DashScope/NVIDIA.

| Variable             | Secret? | Default | Purpose                              |
|----------------------|---------|---------|--------------------------------------|
| MIMIC_IMAGE_ENABLED  | no      | false   | Master switch for mimic draft/render |
| MIMIC_IMAGE_PROVIDER | no      | bfl     | bfl   dashscope   nvidia   openai    |

|                                        |     |                  |                                                               |
|----------------------------------------|-----|------------------|---------------------------------------------------------------|
| BFL_API_KEY                            | yes | —                | BFL FLUX edits when provider=bfl                              |
| DASHSCOPE_API_KEY                      | yes | —                | Alibaba Qwen edit when provider=dashscope                     |
| NVIDIA_NIM_API_KEY                     | yes | —                | NIM Qwen edit when provider=nvidia                            |
| MIMIC_IMAGE_BFL_MODEL                  | no  | flux-2-klein-4b  | BFL model slug                                                |
| MIMIC_VISUAL_SIMILARITY_PCT            | no  | 70               | Reference-edit fidelity hint                                  |
| MIMIC_IMAGE_INPUT_MODE                 | no  | reference_edit   | reference_edit Or analysis_t2i (new visual / invented plates) |
| WHY_MIMIC_REQUIRE_SUBSTANTIVE_SIL      | no  | true             | Planning guard for Why Mimic carousel                         |
| MIMIC_LAYOUT_QA_ENABLED                | no  | true             | Post-render layout QA reprint loop                            |
| MIMIC_LAYOUT_QA_MAX_REPRINT_ITERATIONS | no  | 3                | Max layout QA reprint iterations                              |
| MIMIC_IMAGE_DEFAULT_SIZE               | no  | 1024x1536        | Output dimensions                                             |
| LLM_MIMIC*_MAX_CHARS                   | no  | see .env.example | Mimic prompt context caps                                     |

Full list: `.env.example` (comments) and `docs/MIMIC_FLOWS_COMPLETE_GUIDE.md` §13.

## 11. Duplicate reference

Canonical **comments** for many Next.js vars also live in `.env.example` at the repo root. Prefer updating `.env.example` when adding new app-level vars, and add a row here in the matching section.

*Inventory version: aligned with repo scan 2026-06-15. Update when new `process.env.*` appears in code.*